



# GANESH CHHETRI

## Software Engineer

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Portfolio

### 1. PROFILE

Software engineer with three years of experience developing and supporting enterprise applications in production at NGN Technologies Private Limited, Thimphu. The work covers the full lifecycle: gathering requirements with business users, designing the data model, building the services and screens, deploying to cloud or on-premise infrastructure, and handling support and security work after release. Currently project lead on two systems for government agencies. A recurring part of the work has been integrating Bhutan NDI, the national digital identity, into existing systems. Holds a four-year B.Tech in Computer Science & Engineering with specialisation in Cloud Technology and Information Security (CGPA 8.38, First Division with Distinction), and is an AWS Certified Cloud Practitioner and Certified DevOps Engineer.

### 2. EMPLOYMENT

| Designation                            | Organization                                                    | From     | To       |
|----------------------------------------|-----------------------------------------------------------------|----------|----------|
| Assistant Software Engineer (Grade T1) | NGN Technologies Private Limited<br>e-Centric Division, Thimphu | Aug 2023 | Jan 2024 |
| Software Engineer (Grade S1)           | NGN Technologies Private Limited<br>e-Centric Division, Thimphu | Feb 2024 | Feb 2026 |
| Software Engineer (Grade T2, Band 2)   | NGN Technologies Private Limited<br>e-Centric Division, Thimphu | Mar 2026 | Present  |

Three years of continuous experience since August 2023, entirely on production systems.

### 3. PROJECT EXPERIENCE

| Period         | Project                     | Role                    | Scope and Contribution                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|-----------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 – Present | Mines Administration System | Project Lead            | A 34-module platform for the Department of Geology and Mines covering mining and quarrying leases, transport permits, stockyard balances, royalty calculation and online payment, multi-level approval workflows and role-based access control. Delivered through user acceptance testing and deployed on Kubernetes.                                                                      |
| 2025           | DRUG-RIMS                   | Project Lead            | Registration and licensing of pharmaceutical products for the Bhutan Food & Drug Authority, including a Certificate of Product Registration rebuilt to an official document standard. A security hardening exercise closed OWASP ZAP findings, fixed an authorization bypass, configured content security policy and headers at the proxy, and cleared flagged dependency vulnerabilities. |
| 2024 – Present | Bhutan NDI integration      | Across projects         | Bhutan NDI integrated into DRUG-RIMS, the Mines Administration System, DRA, BCSEA and the G2C Feedback System — credential issuance, proof requests against the citizen identity wallet, passwordless login, webhook registration and event handling over NATS.                                                                                                                            |
| 2024 – 2025    | Digital Health Platform     | Developer               | Application modules for a national health screening programme — dynamic screening, assessment, consent capture, behaviour tracking, email verification and search — with the AWS infrastructure behind them: scheduled Lambda warming, VPC and NAT gateway networking, bulk user provisioning and IAM policy design.                                                                       |
| 2024           | ETool                       | Developer               | Report definition, report generation and the data access layer beneath them.                                                                                                                                                                                                                                                                                                               |
| 2024           | G2C Feedback System         | Support and enhancement | A citizen feedback platform for government-to-citizen services, integrated with Bhutan NDI and using NATS for event handling.                                                                                                                                                                                                                                                              |

|             |                                  |                         |                                                                                                                                                                                                                                                                                                       |
|-------------|----------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024        | BCSEA examination services       | Support and enhancement | Online payment integration for examination services, and optimisation of the views behind the production reporting database.                                                                                                                                                                          |
| 2024        | Food and drug regulatory systems | Support and enhancement | Maintenance and enhancement of live regulatory systems: new features, resolution of long-standing defects, foreign data wrapper links for cross-database reporting, recovery of a medical products database from character-encoding corruption, and correction of CORS configuration across services. |
| 2023 – 2024 | Report Management System         | Developer               | Report definition, generation and export for end users, built in Python.                                                                                                                                                                                                                              |
| 2023 – 2024 | User Management System           | Developer               | Account creation and lifecycle, role definitions and the permission model determining what each role can access, built in Python.                                                                                                                                                                     |
| —           | Document and invoice automation  | Independent work        | Tooling to generate Word documents and invoices for Nimalung Dratshang and Yuvan Infotech, replacing preparation done manually.                                                                                                                                                                       |

#### 4. CORE RESPONSIBILITIES

- **Project delivery** — Lead delivery of two government systems end to end: requirement gathering, technical development plan, development, integration, documentation, user acceptance testing and go-live, coordinating between business units and the technical team.
- **Application maintenance** — Maintain integrated enterprise applications and customer-facing portals in production, covering releases, defect resolution and user support.
- **Application security** — Remediation of OWASP ZAP findings, content security policy and header configuration, closure of an authorization-bypass flaw, dependency vulnerability clearance and proxy hardening.
- **Database administration** — Administration of production PostgreSQL and MySQL databases: backup and restoration, foreign data wrapper links for cross-database reporting, view and query optimisation, and recovery from character-encoding corruption without data loss.
- **Cloud and deployment** — AWS infrastructure for the Digital Health Platform, and operation of the Mines Administration System on Kubernetes with Jenkins pipelines, nginx and Docker.
- **Root-cause analysis** — Diagnosis and resolution of production incidents including an SSO and Keycloak redirect loop, cross-service workflow status divergence and CORS misconfiguration.
- **Stakeholder engagement** — Requirement-gathering sessions and user acceptance testing with business users, translating process requirements into technical specifications and documentation.

#### 5. TECHNICAL SKILLS

|                  |                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages        | Java, JavaScript and TypeScript, Python, SQL, HTML, CSS                                                                                                                 |
| Backend          | Spring Boot, Spring MVC, Spring Security, Hibernate and JPA, REST APIs, Node.js, FastAPI, Django                                                                        |
| Frontend         | Angular, Next.js and React                                                                                                                                              |
| Databases        | PostgreSQL, MySQL, MongoDB; backup and restoration, foreign data wrappers, indexing, view and query optimisation                                                        |
| Cloud and DevOps | AWS (Lambda, API Gateway, EC2, S3, IAM, Cognito, EventBridge, VPC, Secrets Manager), Kubernetes, Docker, Jenkins, nginx, Linux, Git, Maven                              |
| Digital identity | Bhutan NDI — credential issuance, proof requests, passwordless login, webhook registration, NATS event handling                                                         |
| Security         | OWASP ZAP remediation, content security policy and headers, JWT, role-based access control, SSO with Keycloak, dependency scanning, patch management, secure API design |
| Practices        | Requirement gathering, system analysis and design, technical specification, user acceptance testing, production support, project coordination                           |

#### 6. EDUCATION

| Qualification                                                                                                  | Institution                                                                                       | Year        | Result                   |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------|--------------------------|
| Bachelor of Technology (B.Tech)<br>Computer Science & Engineering<br>(Cloud Technology & Information Security) | M. M. Engineering College, Mullana<br>Maharishi Markandeshwar (Deemed to be University),<br>India | 2019 – 2023 | CGPA 8.38<br>Distinction |
| Bhutan Higher Secondary Education Certificate<br>(Class XII, Science)                                          | Samtse Higher Secondary School<br>Bhutan Council for School Examinations and                      | 2018        | 58.4%                    |

|                                                            |                                                                                         |      |              |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------|------|--------------|
|                                                            | Assessment                                                                              |      |              |
| <b>Bhutan Certificate of Secondary Education (Class X)</b> | Samtse Higher Secondary School<br>Bhutan Council for School Examinations and Assessment | 2016 | <b>73.0%</b> |

## 7. CERTIFICATIONS AND HACKATHONS

| Certification / Training / Award                                     | Issuing Body                                       | Date     |
|----------------------------------------------------------------------|----------------------------------------------------|----------|
| <b>AWS Certified Cloud Practitioner (CLF-C02), valid to Sep 2027</b> | Amazon Web Services                                | Sep 2024 |
| <b>Certified DevOps Engineer (CDE)</b>                               | Global Skill Development Council                   | Sep 2025 |
| <b>PMO and Solution Architecture Capacity Development Program</b>    | Accenture Japan and JICA                           | Feb 2025 |
| <b>Web Application and API Protection, Level 1</b>                   | Prophaze Technologies                              | Aug 2023 |
| <b>Introduction to Data Science with Python</b>                      | Omdena Academy                                     | Dec 2024 |
| <b>Participant, Bhutan DevFest International Hackathon</b>           | GovTech Agency, Bhutan NDI and Ethereum Foundation | Jun 2025 |
| <b>Participant, Blockchain-Based Land Registry Hackathon</b>         | GovTech Agency and National Land Commission        | Jun 2025 |
| <b>Participant, DHI and AWS Hackathon</b>                            | Druk Holding and Investments with AWS              | Oct 2024 |

*Referees available on request.*